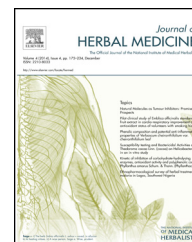


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Review

Natural molecules as tumour inhibitors: Promises and prospects



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ABSTRACT

Numerous natural substances derived from plants have demonstrated promising roles in providing substantial benefit in the prevention, treatment and management of various forms of cancer. Of the myriad of natural constituents found in plants, only a limited number have been screened for their potential role in cancer prevention or their cancer inhibition efficiency using *in vitro* or *in silico* models. Of those constituents that have been screened, an even smaller number are ultimately the subject of further studies in animal models or full scale double blind placebo controlled clinical studies in humans.

This paper comprises a review of the available literature with regard to the potential benefits of natural substances in the prevention, treatment and management of cancer and aims to provide readers with an insight on the potential benefits of the same. The paper will look at the mechanisms of action through altering multiple pathways and the benefits of some of the more comprehensively researched plant derived molecules like resveratrol, damnacanthal and morindone, anthocyanins and garcinol. Details on natural molecules derived from microbial and marine sources, namely didemnin, plitidepsin, dolastatins and bryostatin-1, are also discussed. Furthermore, information on pharmacological and toxicological advantages of natural molecules over synthetic molecules in the treatment of cancer is explored. Finally, the authors consider the means by which to achieve optimum benefits from natural compounds in cancer therapy, along with discussion regarding future approaches to cancer drug discovery.

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1. Introduction

Cancer treatment modalities have been constantly improving year on year, but a comprehensive and uniformly reliable cure for this ailment is still a long way off. Cancer chemotherapeutics can often provide temporary relief of symptoms, increase life expectancy, and occasionally provide prolonged relief or minimize risk factors. In recent years, efforts have been made to identify and synthesize possible potential anticancer molecules/drugs. However, extensive synthetic chemistry research has led to relatively small improvements in the existing prototype drugs. So, with the availability of fewer novel molecules for therapeutic application, modification of existing drugs continues to be an important aspect of cancer therapeutics research. There is a continued need for development of new prototypes, i.e., new templates to use in the design of potential chemotherapeutic agents. Natural products are constantly providing such templates for prophylactic and therapeutic purposes in cancer and other chronic diseases. Recent studies into tumour-inhibiting compounds of natural origin have yielded an impressive array of novel structures and these are capable of acting through multiple pathways to ensure selective killing of cancer cells alone. Many of these natural products are extremely complex in structure, and it is most unlikely that such compounds would have been synthesized in empirical approaches to new drugs. Some of the major challenges with synthetic molecules include, *inter alia*, the complexity of the molecules, stereochemistry of natural products, together with the balancing between toxicity and induction of programmed cell death. In their natural form, bioactive molecules occur in a specific stereochemical formation that is pharmacologically active. These molecules will be either inactive or less efficient when extracted or separated from the plant cellular organization; mainly due to changes in structural conformation. It is also difficult synthetically to produce these molecules in the same manner as they exist in nature. For example, bioavailability of β -carotene from a natural source (which contains both *cis* and *trans* stereoisomers) will be more

biologically active than synthetic β -carotene, which only contains the *trans* isomer (Ben-Amotz and Levy, 1996). Similarly many alkaloids are known to be either less potent or have a different pharmacological effect with changes in optical isomerism.

A plethora of natural compounds has been tested for their cytotoxic potential. One of them is from cucurbitaceous plants containing steroid nucleus phytochemicals, commonly known as Cucurbitacins, which could induce differentiation and arrest cell cycle growth *in vitro* (Haritunians et al., 2008). Unfortunately, their high toxicity *in vivo* resulted in a narrow therapeutic index, making them relatively unpromising therapeutic agents. In addition to this, other molecules have demonstrated their multimodal benefits in cancer prevention and progression. These include resveratrol, anthocyanin, damnacanthal, morindone, garcinol, didemnins, plitidepsin bryostatins and dolastatins. Table 1 summarizes some natural molecules that have made significant contributions towards cancer management. Structures for these highly promising molecules are depicted in Fig. 1. The advantage of considering molecules derived from dietary sources is their relatively low toxicity to normal tissue. This review provides details of some of the promising natural compounds that are effective in both *in vitro* and *in vivo*. Additionally, information on ingredients that are part of various formulations available for cancer management is also provided.

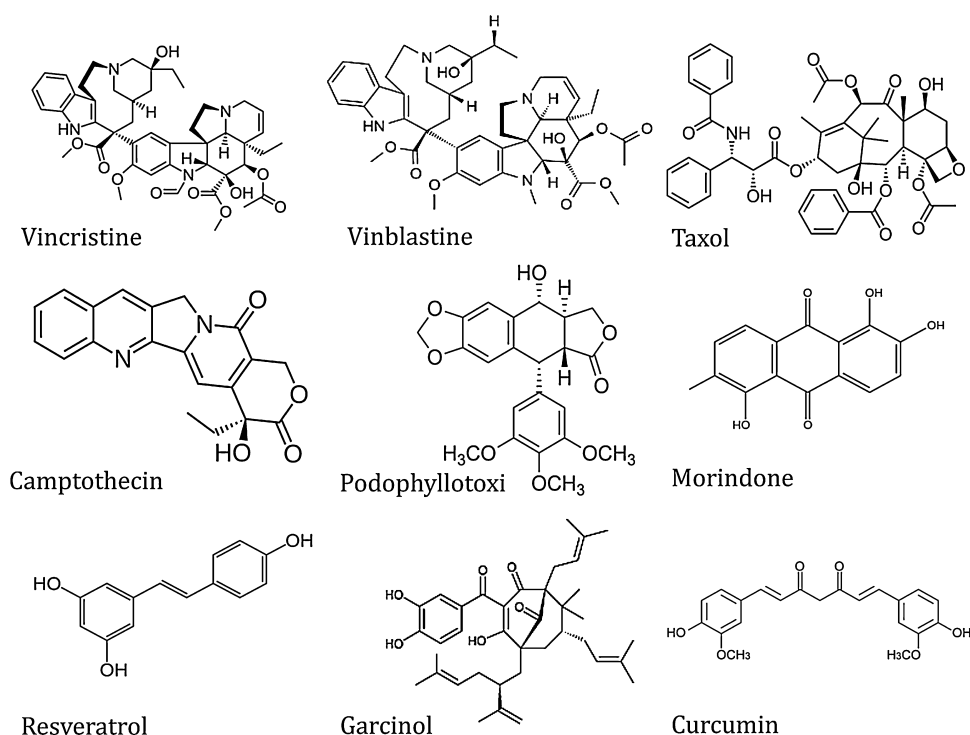
2. Resveratrol

Resveratrol (3,5,4'-trihydroxystilbene) is a naturally occurring dietary polyphenol and phytoalexin mainly found in plants such as grapes (*Vitis vinifera*), peanuts (*Arachis hypogaea*) and mulberries. Resveratrol has been demonstrated to have an interesting spectrum of pharmacological properties. It is indicated in various ailments, such as cardiovascular disorders (Leifert and Abeywardena, 2008), diabetes mellitus, lipid metabolism disorders (Wu et al., 2013), inflammation, cancer (Wu et al., 2013), HIV and bacterial infections (Aggarwal et al., 2004a,b).

Table 1 – Natural molecules that have shown success for cancer treatment and associated symptoms.

Name of the drug	Source of the drug	Mechanism of action	Reference
Vincristine (>99% pure; LD ₅₀ = 1.1 ± 0.1 mg/kg b.w. i.p. rats)	<i>Catharanthus roseus</i>	Binds to tubulin and inhibits its polymerization into microtubules, which prevents spindle formation in mitosing cells and causes arrest at metaphase	Jordan et al. (1991)
Vinblastine (>99% pure; TDLO 557 µg/kg i.p. humans)			
Camptothecin (>98% pure; LD ₅₀ = 64 mg/kg. i.p. mice)	<i>Camptotheca acuminata</i>	Binds to and inhibits topoisomerase I	Liu et al. (2000)
Podophyllotoxin (>97% pure; LD ₅₀ = 30 mg/kg. i.p. mice)	<i>Podophyllum hexandrum</i>	Inhibits mitochondrial function and nucleoside transport. Also has an effect on topoisomerase II	van Maanen et al. (1988)
Morindone (present in extract at µg levels, >2000 mg/kg i.p. Noni extract in rats)	<i>Morinda citrifolia</i>	Acts on the various molecular targets, such as NFκB, p53, COX-2, iNOS	Faltynek et al. (1995), Nualsanit et al. (2011)
Resveratrol (>98% pure; TDLO 1.1 mL/kg oral. humans)	<i>Vitis vinifera</i>	Acts on the various molecular targets such as NF-κB, p53, COX-2, caspases, iNOS	Saiko et al. (2008)
Garcinol (>99% pure; LD ₅₀ = 980 mg/kg b.w. i.p. rats)	<i>Garcinia indica</i>	Acts on the various molecular targets such as NFκB, COX-2, iNOS, HATs and caspases	Pan et al., 2001; Balasubramanyam et al. (2004)
Taxol (>99% pure; LD ₅₀ = 85 mg/kg b.w. i.v. rats)	<i>Taxus brevifolia</i>	Acts on microtubules, stabilizing them in the polymerized state	Kumar (1981)
Curcumin (>99% pure; LD50 > 2000 mg/kg. oral. mice)	<i>Curcuma longa</i>	Induces apoptosis, reduces inflammation and selective cytotoxicity through multiple pathways	Mimeault and Batra (2011)

COX, cyclo-oxygenase; HAT, histone acetyl transferase; iNOS, inducible nitric oxide synthase; LD₅₀, lethal dose in 50% of animals; NF, nuclear factor; TDLO, toxic dose low (minimal toxic dose).

**Fig. 1 – Chemical structures of the major natural molecules with promising anti-proliferative/anti-tumour activity.**

In vitro studies of resveratrol have revealed its efficacy in suppressing proliferation of numerous tumour cells, including lymphoid and myeloid cancers (Roman et al., 2002), breast (Mgbonyebi et al., 1998), colon (Delmas et al., 2003), pancreas (Ding and Adrian, 2002), stomach (Holian et al., 2002), prostate (Hsieh and Wu, 1999), ovary (Opipari et al., 2004), liver (Delmas et al., 2000), and lung (Kim et al., 2003) cancers, in addition to melanoma (Niles et al., 2003). Additionally, its proliferation inhibition effect and anti-mitogenic effect on smooth muscle cells reflects its cardio protective activity (Zou et al., 1999). Resveratrol is also known to modulate several signalling kinases [Raf, Src, mitogen-activated protein kinase (MAPK), protein kinase (PK)D and PKC δ], transcription factors (p53/p21 and 1kB kinase/NF- κ B) and other specific targets (TRAIL/DR4+5, Fas/CD95 and PI3K/Akt) leading to selective cancer cell death (Saiko et al., 2008).

In vitro and *in vivo* studies of resveratrol have demonstrated its safety and efficacy. Resveratrol is efficiently absorbed upon oral consumption in humans, however its bioavailability is limited due to rapid phase II metabolism (Patel et al., 2011). The colon has been a potential therapeutic site for resveratrol, as its absorption is maximal in the intestine (Juan et al., 2012).

Resveratrol is reported to be safe apart from mild to moderate gastrointestinal side effects. Resveratrol-3-O-sulfate, resveratrol-4'-O-glucuronide and resveratrol-3-O-glucuronide have been reported to be the major plasma metabolites (Brown et al., 2010).

The insulin-like growth factor (IGF) signalling system, which consists of IGFs, IGF-binding proteins (IGFBPs) and IGF receptors, crucially influences malignant development. IGFs possess potent anti-apoptotic and mitogenic properties (Ibrahim and Yee, 2004; Butt et al., 1999) and affect cell differentiation, neoplastic transformation and metastasis (Butt et al., 1999; Lopez and Hanahan, 2002). Consumption of resveratrol significantly reduced the levels of IGF-1 and IGFBP-3. In a repeat dose study reported from university of Leicester, UK, 40 healthy volunteers ingested different concentrations (0.5, 1.0, 2.5, or 5.0 g/day) of resveratrol for 29 days. Following this concentration of resveratrol and its metabolites in serum were measured using high performance liquid chromatography between 21 and 28 days. Level of IGF-1 and IGFBP-3 was measured by ELISA technique on day 29th before dosing resveratrol (Brown et al., 2010). This mechanism might also contribute to cancer chemopreventive activity. Some of the key markers altered by resveratrol include, tumour necrosis factor (TNF)- α , p21, p53, survivin, nuclear factor (NF) κ B, MAPK and matrix metalloprotease (MMP)-9; these markers are altered at the dose of 100 mg/kg of body weight or less, indicating the potency of resveratrol (Bishayee, 2009). A phase I clinical study using a single dose of resveratrol in 10 volunteers was carried out: there were no serious adverse reactions observed in any of the patients and a maximum plasma concentration of 539 ± 384 ng/mL was observed. Considering the efficacy of resveratrol at $5 \mu\text{mol/L}$ based on *in vitro* studies, it is recommended that achieving higher bioavailability is essential for achieving organ-specific cancer chemopreventive benefits (Boocock et al., 2007). Further research is essential via either targeted drug systems or nano-formulations.

3. Damnacanthal and morindone

Damnacanthal is an anthraquinone component extracted from the roots of *Morinda citrifolia* L., which has been used as a traditional therapy for several chronic diseases including cancer (Wang et al., 2002). This is more commonly known as 'noni' in different regions. Noni is also reported to have a broad range of health benefits in cancer, infection, diabetes (Wang et al., 2002), asthma (Wang et al., 2002), cough, hypertension, pain, ulcers, wounds, haemorrhoids and rheumatoid arthritis (Wang et al., 2002). Compounds such as scopoletin, rutin, ursolic acid, and β -sitosterol are also considered to be key components of noni (Lv et al., 2011).

Damnacanthal is a Src tyrosine kinase inhibitor reported to have potent and selective p56lck inhibitory effects (Faltynek et al., 1995). It is also associated with inhibition of NF κ B, thus suppressing the procytokines, cyclooxygenase-2 (COX-2), and inducible nitric oxide synthase (iNOS); therefore, it is helpful in treating inflammatory-related disorders (Nualsanit et al., 2011). Damnacanthal has also been reported to induce apoptosis through sustained activation of the p38 MAPK pathway, leading to the transcription of the death receptor family genes encoding DR5/TRAIL, TNF-R1/TNF- α genes, as well as the p53-regulated Bax gene. This is known to activate caspase-8, leading to Bid cleavage. In turn, activated Bid, acting with p53-regulated Bax, leads to cytochrome c release from mitochondria into the cytoplasm. Combined activation of the death receptors and mitochondrial pathways results in activation of the downstream effector caspase 3, leading to cleavage of Poly (ADP-ribose) polymerase, popularly known as PARP (Lin et al., 2011) followed by apoptosis. Damnacanthal is also able to exhibit cell growth arrest in colorectal cancer cells and induce caspase-8 activity (Nualsanit et al., 2012). Damnacanthal and morindone have also shown intensive inhibitory effects against topoisomerase II (Tosa et al., 1998).

Noni juice rich in polysaccharide content has been found to possess both prophylactic and therapeutic potential against the immunomodulatory sensitive sarcoma 180 tumour system. The polysaccharide-rich content could synergise the beneficial effects when combined with a broad spectrum of chemotherapeutic drugs, including cisplatin, adriamycin, mitomycin-C, bleomycin, etoposide, 5-fluorouracil, vincristine or camptothecin (Furusawa et al., 2003).

A study involving the introduction of cell strains of Lewis lung carcinoma to mice, that were selectively bred to be highly susceptible to this type of cancer, reported that the mice treated with the noni juice containing morindone and damnacanthal showed a significant benefit when treated at both therapeutic levels (at a dose of 0.5 mg dissolved in 0.1 mL water which began on day 1 or 2, continued every other day) and prophylactic levels (at a dose of 0.5 mg, i.p. was done on days 5, 3, 2 and 1 before tumour inoculation doses). Following the introduction of tumour cells, after 40 days 31% of the animals survived with therapeutic treatment and 53% with prophylactic treatment, compared to a control in which there was 100% death after 40 days (Hirazumi and Furusawa, 1999).

There are several clinical trials reporting the use of noni for health benefits and the prevention of certain ailments. Upon evaluating haematological and biochemical parameters

obtained from a single-centre, double-blind, three-dose level, parallel group, placebo-controlled safety study, the juice was reported to be safe for consumption (West et al., 2006). Analysis of antioxidant activity of noni juice in heavy smokers (individuals smoking more than 20 cigarettes per day) suggests that it has antioxidant benefits. Consumption of juice for 30 days at 29.5 and 118 mL demonstrated that there was a significant decrease in the plasma superoxide anion radicals (SAR) and lipid hydroperoxide (LOOH) in all patients, again suggesting antioxidant effects (Wang et al., 2009). A phase I clinical trial was conducted using 2–10 g of fruit extract in cancer patients for a period of 28 days. Observations were recorded for toxicity, quality-of-life and pharmacokinetic parameters. The outcomes of the study suggested that there was no toxicity observed as per CTCAE (Common Terminology Criteria for Adverse Events) criteria, significant decrease in the fatigue and pain was observed with treatment of noni and, based on the response, the study was recommended for a phase II trial (Issell et al., 2005, 2009).

4. Anthocyanins

Anthocyanins are a class of polyphenols possessing a broad spectrum of therapeutic and anti-cancerous properties (Bagchi et al., 2004). These molecules usually impart red pigmentation to fruits and are considered as replacements for synthetic colourants for food, pharmaceutical (Neelwarne et al., 2007) and other applications, due to their bright attractive hue and water solubility. These are abundantly found in the fruits of *Garcinia indica* (kokum), banana bracts (Pazmino-Duran et al., 2001), red radish (Giusti and Wrolstad, 1996), strawberry (da Silva et al., 2007), capulin cherry (Ordaz-Galindo et al., 1999), black raspberries (Tian et al., 2006), roselle hibiscus (Tsai et al., 2002), berries (Longo and Vasapollo, 2006) and sweet corn (Jing and Giusti, 2005).

Studies have shown that anthocyanins are beneficial in reducing age-associated oxidative stress, as well as in improving neuronal and cognitive brain function (Shih et al., 2010). Additionally, anthocyanins have been shown to lower the risk of cardiovascular disease, diabetes, arthritis and cancer due to their antioxidant and anti-inflammatory activities (Prior and Wu, 2006). Anthocyanins also provide benefits to diabetics through sensitization of insulin, suppression of α - β -glycosidase and intervention with energy metabolism (Murthy and Manchali, 2012).

Regular consumption of anthocyanins is known to reduce the incidence of colon cancer (Wang et al., 2013). Anthocyanins are metabolized into phenolic acids and aldehydes; gallic acid, 3-O-methylgallic acid and 2,4,6-trihydroxybenzaldehyde. All of these metabolites offer protection against colon cancer (Forester and Waterhouse, 2010). Vascular endothelial growth factor (VEGF), one of the major angiogenesis factors is induced in growing tumours and stimulates endothelial cell proliferation and migration, primarily through the VEGF receptor type 2 (VEGFR2, Flk1/KDR; Ushio-Fukai and Nakamura, 2008). Anthocyanins are also capable of inhibiting VEGF-regulated angiogenesis and inducible NF κ B transcriptions (Bagchi et al., 2004). On the other hand, these could inhibit proliferation and induce apoptosis in human gastric adenocarcinoma

cells through caspase-3 activation (Shih et al., 2005). Cancer chemopreventive activities of anthocyanin can be attributed to their radical scavenging activity, stimulation/activation of phase II detoxifying enzymes, reduced cell proliferation, anti-inflammatory, anti-angiogenic ability and their induction of apoptosis and cell differentiation. Anthocyanins modulate the expression and activation of multiple genes associated with these cellular functions, including genes involved in the PI3K/Akt, ERK, JNK and MAPK pathways (Wang and Stoner, 2008). *In vivo* animal studies have shown that dietary anthocyanins inhibit gastrointestinal (GI) cancers and suppress skin cancer when applied topically (Wang and Stoner, 2008).

A study of 20 individuals demonstrated that supplementing their diets with 300 mg/day of anthocyanins for 3 weeks reduced NF κ B activation. Reduced NF κ B activity was also supported by reductions in interleukin (IL)-8, IL-4 and IL-13 levels; suggesting the role of anthocyanins in chronic inflammation-mediated diseases (Karlsen et al., 2007). The issue with translating the *in vitro*, cell culture and animal studies to human application is serum concentration of the anthocyanins. Anthocyanins have demonstrated influence on cell survival and various oncogenes at concentrations of 10^{-6} – 10^{-4} M. Considering that the maximum serum concentration after a single dose of anthocyanin is 10^{-8} – 10^{-7} M, achieving higher bioavailability is key for utilizing the benefits of anthocyanins in cancer (Cooke et al., 2005).

5. Garcinol

Garcinol is a polyisoprenylated phenolic obtained from the rind of the kokum fruit (*G. indica*). Traditionally these fruits were used in the treatment of various inflammatory ailments, rheumatic pain and bowel complaints (Baliga et al., 2011).

Garcinol is known to induce apoptosis in leukaemia HL-60 cells by triggering the release of cytochrome-c into the cytosol, increasing the activity of procaspase-9, activation of caspase-3 and caspase-2, degradation of PARP and induction of DNA fragmentation by caspase-activated deoxyribonuclease through the digestion of DFF-45 (Pan et al., 2001).

Histone acetyltransferases (HAT) and deacetylases (HDACs) are key regulatory steps for gene expression and cell cycle. Functional abnormalities in these enzymes may lead to cancer (Balasubramanyam et al., 2004). Hence the HATs and HDACs are potential targets for cancer chemotherapy. Garcinol is reported to be potent inhibitor of HATs both *in vitro* and *in vivo* (Balasubramanyam et al., 2004). It has also been demonstrated to be a potent apoptosis inducer and to alter the global gene expression in HeLa cells (Balasubramanyam et al., 2004).

Garcinol has demonstrated induction of apoptosis in breast cancer cells (ER-positive MCF-7 and ER-negative MDA-MB-231; Ahmad et al., 2010), prostate (LNCaP, C4-2B and PC3) and pancreatic (BxPC-3) cancer cells (Ahmad et al., 2011) by down-regulating the NF κ B signalling pathway. Garcinol is also known to have translational potential as a chemopreventive or therapeutic agent against multiple cancers and inhibition of the STAT-3 signalling pathway is one of the mechanisms by which garcinol exerts its anticancer effects (Ahmad et al., 2012).

The 5-lipoxygenase (5-LOX) pathway is known for its association with oral carcinogenesis. Leukotrienes derived from

5-LOX are able to initiate and maintain inflammatory and allergic reactions (Koeberle et al., 2009). The 5-LOX is also overexpressed in various cancer cells (e.g., prostate, breast, lung and colon; Koeberle et al., 2009). Garcinol, a potent inhibitor of the 5-LOX pathway is known to have chemoprotective effects against 7,12-dimethylbenz(a)anthracene (DMBA)-induced hamster cheek pouch carcinogenesis. This was found to be via the interactions between phenolic hydroxyl groups and non-haeme catalytic iron. Treatment of garcinol has been shown to suppress leukotriene B₄ (LTB₄) biosynthesis and inhibit inflammation and cell proliferation in the oral epithelium (Chen et al., 2012).

TNF-related apoptosis-inducing ligand (TRAIL), discovered in 1995, was heralded as a selective killer of tumour cells, and it is still thought to have good potential. TRAIL has been shown to interact with the cell surface through five distinct receptors, namely death receptor (DR)4, DR5, decoy receptor (DcR)1, DcR2 and osteoprotegerin (Aggarwal et al., 2004a,b). Garcinol enhanced TRAIL-induced activation of caspase-8, -9 and -3, thus leading to enhanced PARP cleavage and suggesting that it could enhance the TRAIL-induced apoptosis in HCT116 colon cancer cells (Prasad et al., 2010). Down-regulation of anti-apoptotic proteins survivin, bcl-2, XIAP and both forms of cFLIP is another mechanism by which garcinol could potentiate TRAIL-induced apoptosis (Prasad et al., 2010).

Due to poor oral bioavailability and rapid metabolism clearance from the body, garcinol's complete biological benefits are not utilized. To overcome this, substituted fluorinated 2'-hydroxy chalcones that are structurally similar to garcinol were synthesized and found to be effective against human pancreatic BxPC-3 cancer cells, which possess up-regulated COX-2 expression. These molecules also showed activity against human breast cancer BT-20 cells with triple negative phenotype, suggesting that these compounds will have broader applications (Padhye et al., 2010). Panaxanthone, consisting of approximately 80% α -mangostin and 20% γ -mangostin isolated from the pericarp of the *Garcinia mangostana* L. plant, has demonstrated mammary tumour inhibition activity and anti-metastasis activity in BALB/c mice administered with BJMC 3879 cells. Supplementing with 2500 and 5000 ppm (parts per million) of panaxanthone in the diet for up to 8 weeks significantly reduced the number of metastases in both lung and lymph nodes (Doi et al., 2009). A clinical study of *Garcinia cambogia* extract equivalent to 500 mg of hydroxyl citric acid along with L-carnitine (600 mg/day) in 35 healthy volunteers for 8 weeks demonstrated a reduction in oxidative damage. Treatment resulted in reductions in both lipid peroxides and the physical symptoms of oxidative stress such as, tired eyes, blurry eyes, muscle pain/stiffness, early satiety, epigastralgia, dizziness, arthralgia and fatigue (Yonei et al., 2008). A number of clinical studies have been performed using *Garcinia* for obesity and associated symptoms (Vasques et al., 2008). However, there are not many clinical studies to support its role in relation to cancer.

6. Metabolites from microorganisms as antitumor compounds

The search for new anti-tumour compounds in nature is by no means only confined to plants. Microorganisms, widely

employed as sources of antibiotics, produce a considerable number of metabolites with anti-tumour activity. Amongst various agents used in the treatment of cancers, dactinomycin, bleomycin, doxorubicin, daunorubicin and mitomycin C are most common. Microorganisms have advantages over plants as far as ease of culture and the opportunity for genetic manipulation are concerned; however, their effect on normal cells and host immunity remains critical for evaluation of benefits. Among different molecules isolated, characterized and studied, the following appear the most promising. Table 2 summarizes some of the chemical research into these marine/microorganism derived molecules.

7. Didemnins

Didemnin B is a depsipeptide isolated from the Caribbean sea-squirt *Trididemnum solidum*. Didemnin was the first drug from marine source to be tested clinically in humans (Xu et al., 2012). This compound has antiviral and potent immunosuppressive properties. It is one of the most potent naturally occurring cytotoxic agents as tested in vitro and also a potent antitumor compound. Didemnin is known to inhibit the growth of L1210 leukaemia cells by 50% when treated at 0.001 μ g/mL (Chun et al., 1986). It has also demonstrated antiproliferative activity against cancers of the prostate, lung and brain, as well as lymphomas. Didemnin B is known to inhibit the synthesis of RNA, DNA and proteins (EF-1 α ; Beidler et al., 1999; Suarez-Jimenez et al., 2012). A phase I study of 53 patients revealed that Didemnin B at doses of between 0.4 and 2.5 mg/m²/week did not cause haematological toxicity. Out of the 53 patients included in the study, hypersensitivity reactions were observed in nine, which was attributed to Cremophor EL that was used as a solvent (Maroun et al., 1998; Gelderblom et al., 2001). Based on these results, a phase II study was performed using 4.3–6.3 mg/m² of Didemnin B in 16 patients with glioblastoma multiforme. No anti-tumour activity was observed and based on these results, further studies into Didemnin B to treat glioblastoma multiforme were not recommended (Mittelman et al., 1999).

8. Plitidepsin

Plitidepsin is a cyclic depsipeptide originally isolated from the Mediterranean tunicate *Aplidium albicans*. Plitidepsin has been proven to be active at nanomolar concentrations against various human malignant cell lines and tumour specimens, such as breast, melanoma and non-small-cell lung cancers (Depenbrock et al., 1998). The primary mechanism for plitidepsin cytotoxicity has not been fully elucidated, but proposed mechanisms include induction of oxidative stress resulting in cellular apoptosis (Garcia-Fernandez et al., 2002), induction of apoptosis via activation of the Rac1-JNK pathway (Garcia-Fernandez et al., 2002; Munoz-Alonso et al., 2008; Cuadrado et al., 2003) and increased levels of cell membrane phospholipid oxidation and DNA oxidation (Gonzalez-Santiago et al., 2006). Plitidepsin also has anti-angiogenic properties characterized by inhibition of gene expression for VEGF and its receptor VEGFR-1 in preclinical

Table 2 – Clinical research into marine/microorganism-derived molecules.

Molecule	Source	Type of cancer	Comments	Reference
Dolastatin 10	<i>Dolabella auricularia</i>	Breast cancer (advanced stage)	Acceptable toxicity, minimal activity in advanced breast cancer	Perez et al. (2005)
Dolastatin 10	<i>Dolabella auricularia</i>	Soft tissue sarcoma	Haematological and vascular toxicity, no GI, hepatic or renal toxicity. Not recommended for soft tissue sarcoma	von Mehren et al. (2004)
TZT-1027	Synthetic derivative of dolastatin 10	Advanced refractory cancer	The most common toxicities were fatigue, anorexia, alopecia, nausea, constipation, leukopenia and neutropenia	Schoffski et al. (2004)
TZT-1027	Synthetic derivative of dolastatin 10	Non-small cell lung cancer	Common toxicities included constipation, anorexia, alopecia, nausea, leukopenia and neutropenia	Horti et al. (2008)
LU 103793	Synthetic analogue of dolastatin15	Metastatic breast cancer	Asthenia, stomatitis, myalgia, increase in serum bilirubin	Kerbrat et al. (2003)
Bryostatin 1	<i>Bugala neritina</i>	Refractory non-haematological malignancy	Conventional toxicities included myalgias, fever, anaemia, fatigue, phlebitis and headache	Grant et al. (1998)
Bryostatin 1	<i>Bugala neritina</i>	Relapsed NHL and CLL	Maximum-tolerated dose was found and it proceeded to phase II trials	Varterasian et al. (1998)
Bryostatin 1	<i>Bugala neritina</i>	Children with refractory solid tumours	Myalgia, photophobia were dose-limiting toxicities, proceeded to phase II trials	Weitman et al. (1999)
Bryostatin 1	<i>Bugala neritina</i>	Advanced renal cancer	Myalgia, fatigue, nausea, headache, vomiting, anorexia, anaemia and lymphopenia were the commonly reported side effects	Madhusudan et al. (2003)
Plitidepsin	<i>Aplidium albicans</i>	Relapsed or refractory NHL	Showed clinical activity in relapsed/refractory T-cell lymphoma	Ribrag et al. (2012)
Plitidepsin	<i>Aplidium albicans</i>	Second line therapy in patients with advanced malignant melanoma	Showed a minor degree of anti-tumour activity	Eisen et al. (2009)
Plitidepsin	<i>Aplidium albicans</i>	Second line therapy for small cell lung cancer	Absence of anti-tumour activity	Eisen et al. (2009)
Didemnin B	<i>Carribean tunicate</i>	Non-small cell lung cancer	Neuromuscular toxicity is dose limiting	Shin et al. (1991)
Didemnin B	<i>Carribean tunicate</i>	Previously treated epithelial ovarian cancer	Ineffective	Malfetano et al. (1993)

CLL, chronic lymphocytic leukaemia; GI, gastrointestinal; NHL, non-Hodgkin's lymphoma.

models ([Broggini et al., 2003](#); [Straight et al., 2006](#); [Biscardi et al., 2005](#)). A phase I clinical study of plitidepsin in people with advanced solid tumours at a dose of 0.133–3.6 mg/m²/week suggests that both muscle and liver toxicity were dose limiting at 3.6 mg/m²/week ([Izquierdo et al., 2008](#)). Based on this, a phase II study was performed using 5 mg/m² as a 3-h i.v. infusion every 2 weeks either alone or with dexamethasone in relapsed and refractory multiple myeloma. Results indicated limited but reproducible activity of the compound in relapsed/refractory multiple myeloma patients. Both plitidepsin and a combination of plitidepsin and dexamethasone were well tolerated in the dose used for the study ([Mateos et al., 2010](#)).

9. Dolastatins

Dolastatins are linear dipepsidides isolated from the sea hare *Dolabella auricularia* and have been found to have cytotoxic effects on various tumour cells. Considering the very small amount derived from the sea hare, efforts were made for

total synthesis, which was accomplished in 1989 ([Pettit, 1997](#)). Dolastatin 10 (a natural peptide known for its cytotoxic effect through antimitotic activity) inhibits microtubule assembly and tubulin polymerization, thereby leading to accumulation of cells in the mitotic state. It binds to tubulin-b at a site close to the vinca alkaloid binding site ([Bai et al., 1990](#)). The high antimitotic activity is partly due to its prolonged cellular retention, which facilitates tubulin binding ([Bai et al., 1990](#)). Dolastatins have also been shown to induce apoptosis in various tumour cell lines, including breast cancer, lung cancer, leukaemias and lymphomas ([Sato et al., 2007](#); [Kalemkerian et al., 1999](#)).

The mechanism of apoptosis induction is most likely to be up-regulation of the pro-apoptotic molecule Bax and the concurrent down-regulation of the anti-apoptotic molecule Bcl-2 ([Kalemkerian et al., 1999](#); [Wall et al., 1999](#)).

Due to their growth inhibition and apoptotic effects on tumour cell lines, dolostatins have been studied in phase I and II clinical trials, which unfortunately failed. One of the derivatives of dolostatin 10, TZT-1027, which differs from dolastatin 10 by the replacement of the terminal dolaphenine

amino acid residues with a phenylamine group, showed potent anti-tumour activity in a murine cancer model and is under evaluation in phase I trials (Kobayashi et al., 1997). Conjugation of dolastain 15 with trastuzumab has demonstrated a potent antibody-drug conjugate to inhibit Her2 positive cells both *in vitro* and *in vivo* (Gianolio et al., 2012).

10. Bryostatin 1

The bryostatins are a group of 20 novel macrocyclic lactones originally isolated from *Bugala neritina*. Bryostatins bind to the regulatory domain of the protein kinase C (PKC) for which they compete with the natural PKC ligands, such as phorbol esters (Wender et al., 1988). Post binding, bryostatin 1 induces the activation of PKC by autophosphorylation and translocation to the cell membrane (Isakov et al., 1993). Bryostatin 1 has been reported to have a variety of effects on tumour cell lines. It has been shown to inhibit proliferation, induce final differentiation and apoptosis as well as showing antineoplastic effects and immunomodulatory activity (Kortmansky and Schwartz, 2003).

As a single compound it has weak apoptotic effects, but in combination with other antitumour drugs, such as paclitaxel, it synergistically increases the paclitaxel-induced up-regulation of caspases, even in Bcl-xL overexpressing cells (Wang et al., 1999). Therefore, it is an interesting agent to be considered for combination therapy.

11. Advantages of naturally derived molecules over synthetic molecules in cancer treatment

For a number of reasons, natural products are likely to continue to provide sources for new commercially viable anticancer drugs. Firstly, there is greater chemical novelty amongst natural resources than in chemicals from any other source (Harvey, 2000). Secondly, natural products can be a more economical source of diverse chemicals than the synthesis of equivalent numbers of chemicals (Harvey, 2000). Thirdly, natural products that are biologically active are generally small molecules with drug-like properties. That is, they are capable of being absorbed and metabolized by the body. Therefore, the development costs of producing orally active medicines are likely to be much lower than with biotechnological products or with most compounds produced to date from combinatorial chemistry. In cancer drug discovery over the time frame of nearly 70 years, out of the 155 small molecules that have been developed, 73% are non-synthetic, with 47% actually being either natural products or directly derived from nature (Newman and Cragg, 2007).

There is growing interest in samples from marine actinomycetes (Subramani and Aalbersberg, 2012) and cyanobacteria (Tan, 2007), particularly with the discovery of species unique to the marine environment and the demonstration that they can produce chemically novel bioactive metabolites.

12. Semi-synthetic/modified natural molecules for better efficacy

The modification of natural drugs is often essential to improve their efficacy as medicines. This is evidenced by the number of natural molecules currently used as drugs following modification to increase their pharmacokinetic or pharmacodynamic properties. Considering this, modification of natural molecules has a highly promising future in cancer drug discovery.

13. Conclusion

Components from natural sources, either plants or microorganisms can interfere with multiple cell-signalling pathways to inhibit cancer cells selectively. Promising chemicals are subsequently tested against a range of standard and established experimental neoplasms, and then considered for preclinical toxicological studies. If these results are sufficiently encouraging, then a large amount of the raw material will be required for further studies. Large-scale extractions and fractionation of the compound along with high-throughput screening tools are essential to meet such requirements.

All the phytoconstituents or their derivatives discussed in this article have made their way into either phase I or II trials as a main constituent in the treatment of cancer or as an adjuvant with chemotherapy. But whether a plant-derived natural product or derivative should be considered worthy of development as a drug is up for question. The availability of plants/natural sources becomes critical for such applications. Some of the more hopeful approaches to this problem include, mass cultivation of plants, enhanced secondary metabolites through tissue culture and other biotechnological interventions, and identification of plants species that contain in larger quantities the molecules or compounds that may be of clinical interest. The latter option is most usually found either in plants of the same genus or closely related species from the same family. Such alternative plants have often been documented in Ayurveda, folk medicine and ancient texts. However, the extremely complex structures of most bioactive natural products frequently preclude satisfactory commercial syntheses.

In the case of microorganisms, they have a particular advantage over plants as far as ease of culture and opportunity for genetic modification is concerned. However, selection of specific microorganisms, their identification, isolation of constituents and potential intoxicants poses considerable challenges and limitations.

While all these naturally derived molecules are pharmacologically active and safe in most situations, one of the concerns commonly expressed is the lack of bioavailability. The research data suggest the need for further evaluation of the bioavailability of compounds before undertaking evaluation of their pharmacological properties. Most modern medicines clinically available for cancer treatments are expensive and known for their toxicity and serious adverse reactions. Therefore, it is important to provide scientific evidence for the benefit of bioactive constituents used in ancient systems of

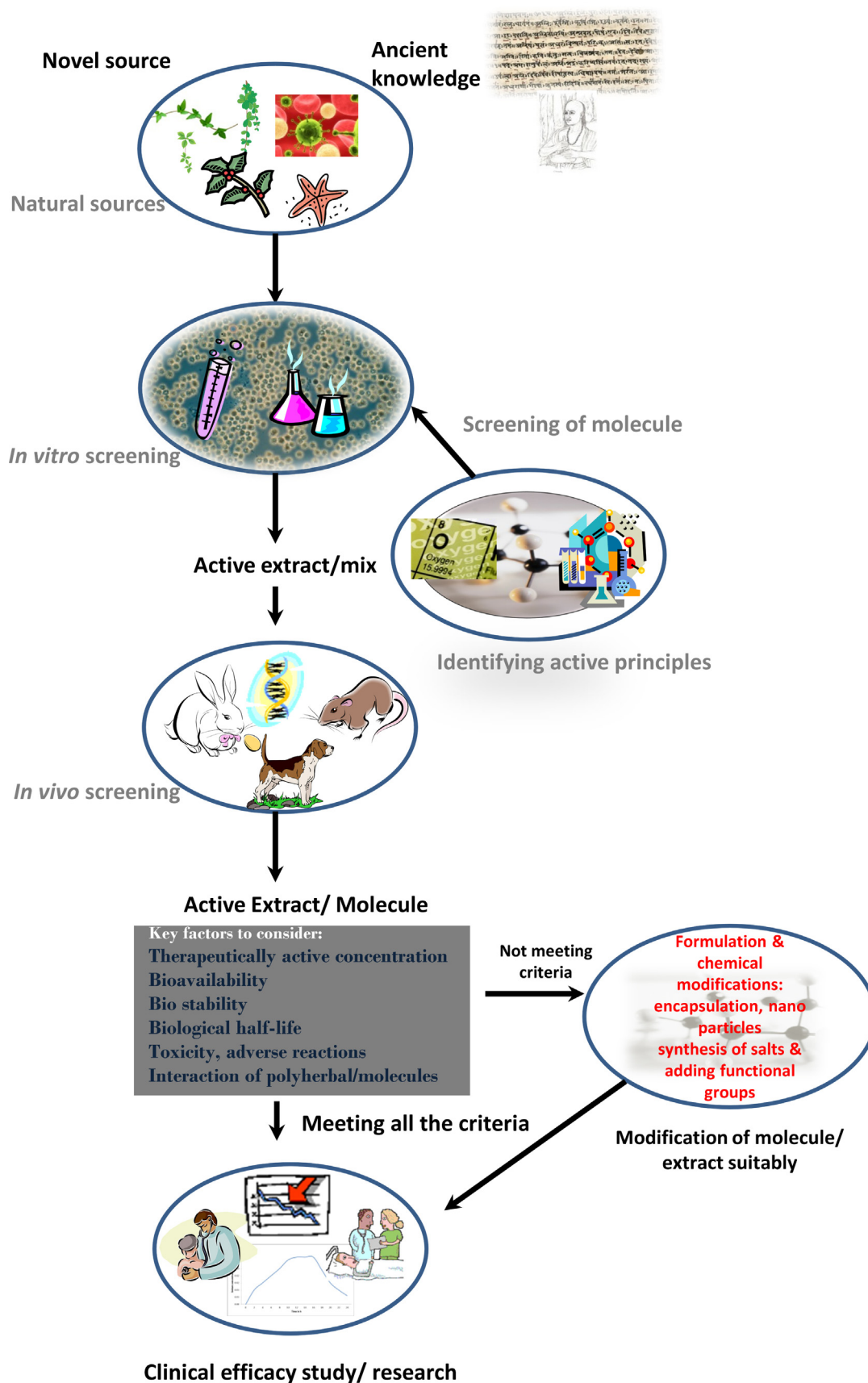


Fig. 2 – Scheme of approach for the identification and development of promising molecules for cancer from natural sources.

medicine and identify novel compounds/molecules from natural sources. More emphasis should be placed on enhancing the efficacy of such natural compounds through increasing the bioavailability and bio-efficacy so that small quantities can be utilized for optimum benefits. Schematic representation of research on effective natural molecules for cancer treatment is depicted in Fig. 2.

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